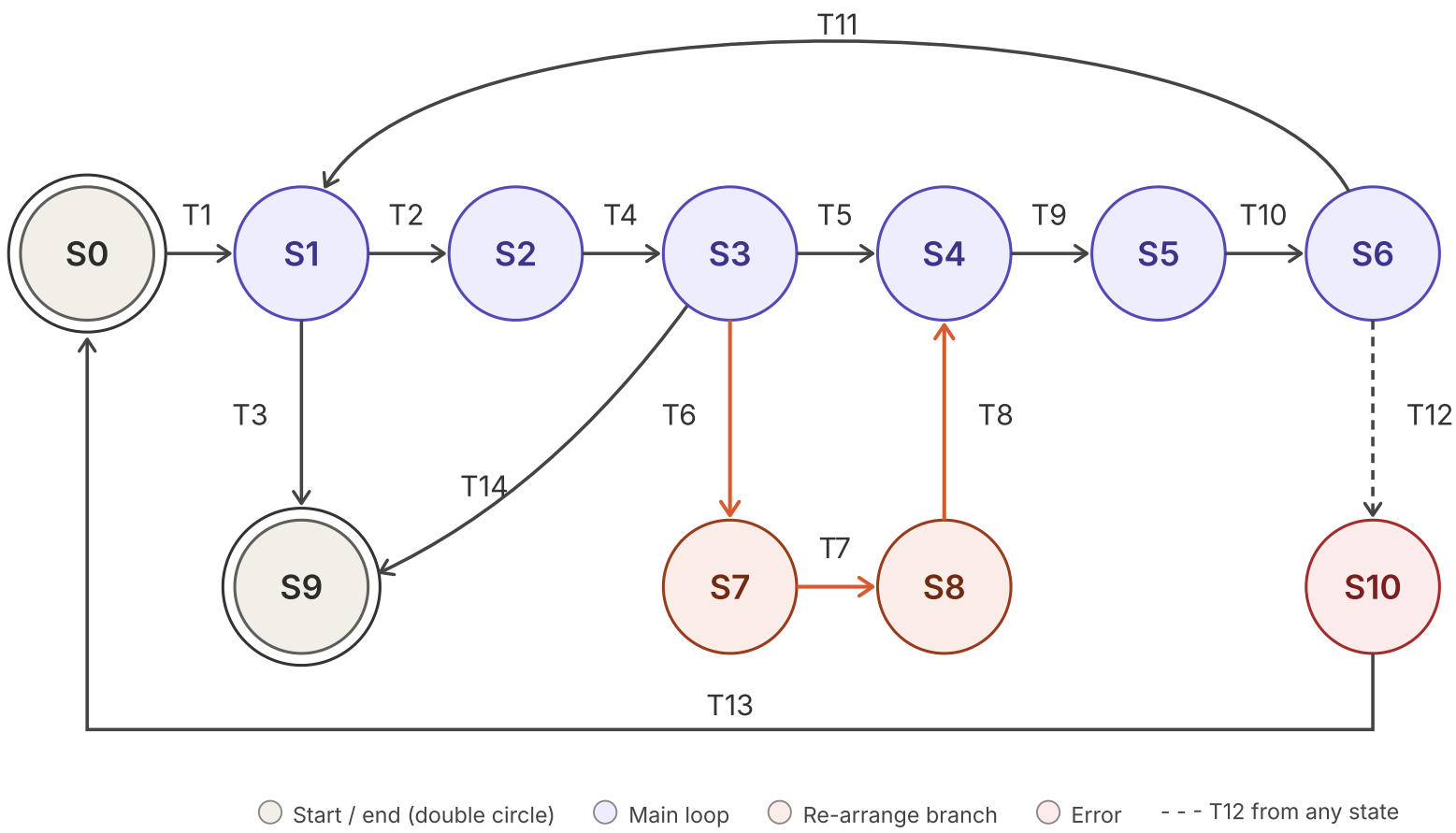
IP12 palletising state machine

Camera detects cubes, robot places them on the pallet, and re-arranges when a new cube fits a used slot better



State definitions

State	Name	What happens
S0	Idle	Robot in home position, waiting for the start signal
S1	Detect	Camera + YOLO find all cubes on the table and read the pallet status
S2	Select	Choose the best cube: highest priority (color/QR), then heaviest
S3	Plan	AI scores every slot and checks the WP4 rules; decides the target slot and whether a swap is needed
S4	Pick	Gripper picks the new cube from the table
S5	Place	Robot puts the cube into the planned slot
S6	Verify	Camera checks the cube is in the right slot
S7	Relocate pick	Robot picks an already placed cube from the better slot (only if nothing is on top of it)
S8	Relocate place	Robot moves that cube to another free slot, freeing the better slot
S9	Done	Pallet full or no cubes left (end state)
S10	Error	Failure: overweight, dropped cube, failed verify; operator help needed

Transitions

ID	From → To	Condition
T1	S0 → S1	Start signal received
T2	S1 → S2	Cubes detected and a free slot exists
T3	S1 → S9	No cubes left or pallet full
T4	S2 → S3	Best cube chosen (weight ≤ 4.0 kg)
T5	S3 → S4	Best slot is free and passes all WP4 rules
T6	S3 → S7	Best slot is taken, swap improves the score by ≥ 2.0, the placed cube was not moved before, fewer than 3 moves so far
T7	S7 → S8	Placed cube is gripped
T8	S8 → S4	Cube moved to a free slot; the better slot is now free
T9	S4 → S5	New cube is gripped
T10	S5 → S6	Cube released in its slot
T11	S6 → S1	Placement confirmed; detect the next cube
T12	any → S10	Failure: overweight cube, gripper lost the cube, verify failed, or robot fault
T13	S10 → S0	Operator fixed the problem and reset the robot
T14	S3 → S9	No valid slot left for any cube